Revision — 1.3.3 Networks

OCR H446 — one-page revision summary

Must know

- **LAN**: small area, single site, organisation owns the hardware, high speed. **WAN**: large area (cities/countries), uses third-party infrastructure; the internet is the largest WAN.
- **Client-server**: a server provides services to clients — central security/backup, scales well, but the server is a single point of failure and costlier. **Peer-to-peer (P2P)**: all peers are equal (act as client and server) — cheaper and no single point of failure, but harder to manage/secure.
- **Internet structure**: *backbone* (high-capacity routes), *ISP* (provides access), *IP address* (uniquely identifies a device; IPv4 = 32-bit, IPv6 = 128-bit), *DNS* (translates domain names → IP addresses, hierarchical and cached).
- **Topologies**: **Star** — devices connect to a central switch/hub; a single cable fault is isolated, but the central node is a single point of failure. **Mesh** — multiple paths between nodes; resilient/fault-tolerant, no single point of failure, but expensive (wired).
- **Packet switching**: data split into packets, routed independently, reassembled at the destination — efficient and resilient (used by the internet). **Circuit switching**: a dedicated reserved path for the whole call — guaranteed bandwidth but wastes idle capacity (phone).
- **Protocol** = rules governing communication; **standard** = agreed spec ensuring interoperability.
- **TCP/IP stack** (data goes *down* when sending, *up* when receiving; each layer adds/removes a header): **Application** (HTTP, HTTPS, FTP, SMTP, POP3, IMAP, DNS) → **Transport** (splits into packets, ports, reassembly; TCP/UDP) → **Network/Internet** (adds source/dest IP, routes; IP) → **Link/Data link** (physical connection, adds MAC; Ethernet, Wi-Fi).
- **Protocols**: HTTP = web pages; HTTPS = encrypted HTTP (TLS/SSL); FTP = file transfer; SMTP = *sending* mail; POP3 = retrieve mail (downloads & removes); IMAP = retrieve mail (stays on server, syncs).
- **Hardware**: NIC (holds MAC address); **switch** connects devices *within* a LAN using MAC; **router** connects *different* networks using IP; WAP gives wireless access to a wired network. **Security**: firewall (packet filtering by rules), proxy (intermediary — hides IP, caches, filters/logs), encryption (protects data in transit).

Key terms

Term	Definition
LAN / WAN	Small single-site network / large geographically spread network
Client-server	Central server provides services to client devices
Packet switching	Data split into packets routed independently and reassembled
Protocol / Standard	Rules governing communication / agreed spec for interoperability
DNS	Translates domain names into IP addresses
MAC address	Unique 48-bit hardware address of a NIC (link layer)
Switch / Router	Connects devices in a LAN (MAC) / connects different networks (IP)
Proxy server	Intermediary that hides IP, caches pages and filters/logs traffic

How to — trace a secure web request

1. **DNS** translates the typed domain name into the server's IP address (queries DNS servers; results may be cached).
2. The browser (client) sends the request to that IP over **HTTPS**, so data is encrypted in transit.
3. At the **transport** layer the data is split into **packets**, each given a header (sequence number, ports).
4. Each packet is given **source/destination IP** addresses and **routed independently** across the network, then reassembled in order at the destination.
5. A **router** forwards the packets between networks toward the destination using IP addresses.

Exam tips

- **Switch** = devices within a LAN (uses MAC); **router** = connects different networks (uses IP). Never swap them.
- Name the TCP/IP layer *and* what it adds: IP at Network/Internet, MAC at Link, ports/packets at Transport.
- Remember two meanings of redundancy — data (bad) vs hardware/backup (good) — this crosses over with 1.3.2.
- SMTP *sends* mail; POP3/IMAP *retrieve* it (POP3 removes from server, IMAP keeps and syncs).